Additive Manufacturing-Based Artificial Fog Collection System to Increase Water Condensation Efficiency – A Multi bioinspired Approach

Literature Review on Design Parameters, Numerical Data for Bioinspired Fog Collectors: Optimizing Water Collection Efficiency

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1. 3D-Printed Cactus-Inspired Spine Structures for Highly Efficient Water Collection [1]

Numerical data and design parameters

Parameter/Dimension	Value	Description
Water Collection Rate	2 mg/min/mm^3	Maximum water collection rate achieved by spines with 10° tip angle.
Tip Angle of Spines	10°, 20°, 30°, 40°, 50°	Different tip angles were tested for optimal water collection.
Best Performing Tip Angle	10°	Achieved the highest water collection rate.
Humidity for Testing	95%	Relative humidity was used in the water collection tests.
Spine Height	1 mm, 1.5 mm	Heights of cactus-mimetic spines.
Optimal Tip Angle for Fast Water Transport	10° (for 1.5 mm spine height)	Optimal angle for rapid water transport along the spine.
Pore Size of Water Droplets	8–40 µm	Size range of water droplets intercepted by the spines.
3D Printing Process	ISA-3DP (Immersed Surface Accumulation 3D Printing)	Used for fabricating microscale biomimetic multibranched spine structures.
Material for 3D Printing	E-glass/MWCNT composite resin	Photocurable resin with multiwall carbon nanotubes (MWCNTs)
Coating for Wettability Modification	Superhydrophobic nano-coating	They are applied to improve water collection efficiency by reducing surface tension.
Surface Roughness Improvement	The stair-stepping effect from 3D printing layers	It helps in collecting water droplets at the edges of spines.
Arrangement of Spines	Hexagonal	Improved airflow and water collection efficiency.
Exposure Time per Layer	Calibrated based on the 2D patterned light beam	Ensures sharp tip creation with dimensions smaller than 8 µm.

Reference:
https://drive.google.com/file/d/13TZnHVIX2JsWCXRTGbz__hsVYf3hDPCh/view?usp=sharing

2. Multi bioinspired Hybrid Superwetting Surface for Efficient Fog Collection and Power Generation [2]

Numerical and Design Parameters

Parameter	Value/Specification	Description
Fog Collection Rate (WCR)	23 g cm^2 h^1	Efficiency of fog collection achieved by the Hybrid Wetting Rod (HWR).
Maximum Voltage	200 V	Generated by the droplet-based electricity generator (DEG).
Maximum Current	2.47 μA	Measured during the electricity generation from fog droplets.
Fog Flow Velocity	50 cm/s	Simulated inlet fog flow velocity used for fluid dynamics simulations.
Contact Angle (SHB surface)	>160°	The superhydrophobic surface contact angle of the HWR.
Sliding Angle (SHB surface)	3°	Measured to assess droplet sliding efficiency.
Tilt Angle of Boomerang-shaped Patterns (θ)	30°	Optimal tilt angle for maximizing fog collection efficiency.
Vertical Distance Between Patterns (h)	50 mm	Distance from the humidifier nozzle to the HWR for fog collection.
Microchannel Width (l)	23 μm	Optimal width of the superhydrophobic microchannel for efficient droplet transport.
HWR Rotation Angle	360°	Full rotation for omnidirectional fog collection in the HWR structure.
Humidity level	90-95%	Optimal fog collection efficiency

Manufacturing process and materials

Step	Description
Material Used for HWR	101 copper rod (ϕ = 1 mm) coated with Cu (OH)2_22 and dodecanethiol for superhydrophobic properties.
Surface Treatment	The HWR surface was laser-etched to create boomerang-shaped patterns mimicking cactus and desert beetle structures.
Laser Power	15 W
Laser Scanning Speed	500 mm/s
Solution Immersion	The copper rod was immersed in a solution of 1 M NaOH and 0.05 M $(NH_4)_2S_2O_8$ for 30 minutes to form a $Cu(OH)_2$ layer and modified with ethanol solution for superhydrophobicity.
Materials for Electricity Generator	The DEG was made of an acrylic substrate, copper foil electrodes, and a fluorinated ethylene propylene (FEP) membrane as the dielectric layer.

Reference:
https://drive.google.com/file/d/1J7bzPQYvLxQB8rnMr8Ipb75kzkauia9t/view?usp=sharing

3. Rapid 3D Printing of Bioinspired Hybrid Structures for High-Efficiency Fog Collection and Water Transportation [3]

Numerical data and design parameters

Parameter	Value	Description	Application in FEA
Spine Height (h)	4.47 mm	Height of cactus-inspired spines.	Structural analysis under wind and fog load.
Base Radius (r0)	0.25 mm	The radius of the spine base.	Mechanical stability analysis.
Apex Angle of Spine	6.4°	The angle at the spine's tip for fog collection.	Aerodynamic efficiency in fog capture.
Number of Longitudinal Ridges (n)	0, 2, 4, 8	Various ridge configurations to enhance fog collection.	Simulating droplet capture efficiency.
Ridge Apex Angle (a)	30°	The apex angle of longitudinal ridges.	Analyzing surface roughness effects on droplet sliding.
Ridge Height (r1 - r0)	0.25 mm	Height of ridges at the spine's base.	Impact on water transport.
Spacing Between Spines (d)	1.33 mm	The center-to-center distance between adjacent spines.	Optimization of layout for fog capture.
Environmental Humidity (RH)	65-90%	Humidity levels used during testing.	Effect of different humidity levels on fog collection.
Transportation Channel Width (C-IAPGs)	0.2 mm	Width of water transportation channel at the base.	Flow analysis of water transport efficiency.
Transportation Channel Depth	0.1 mm	Depth of the C-IAPGs channel for water transport.	Stress and flow simulation for droplet movement.
Water Transport Speed (C-IAPGs)	0.19 mm/s	Speed of water droplet transport in C-IAPGs channels.	Droplet dynamics and transport analysis.
Maximum Build Area (3D Printing)	8.83 x 5.52 mm	Maximum printing area in the 3D printer used for creating samples.	CAD design constraints.
Layer Slicing Thickness (3D Printing)	5 µm	The thickness of each printed layer.	Accuracy in CAD and manufacturing simulation.

Manufacturing Process and Material

Process	Description
3D Printing Technology	Micro-continuous liquid interface printing (µCLIP).
Material for 3D Printing	Photocurable resin with 1,6-hexanediol diacrylate (HDDA).
Resolution	6.9 µm/pixel at the focal plane.
Printing Speed	Up to 125 µm/s for large structures, 12.5 µm/s for accuracy.
Photoinitiator	Phenylbis (2,4,6-trimethylbenzoyl) phosphine oxide.

Reference:
https://drive.google.com/file/d/1By91Ast1oODR7NbOis90doNTleAEll45/view?usp=sharing

4. Bioinspired slippery asymmetric bumps of candle soot coating for condensation and directional transport of water [4]

Numerical data and design parameters

Parameter	Value	Description	Application in CAD/FEA
Bump Height	53.6 nm	Height of asymmetric soot particle bumps.	Structural analysis of bump shape for water transport.
Bump-to-Bump Distance	2.59 µm	Distance between adjacent soot particle bumps.	Optimization for droplet nucleation and transport.
Peak Width	0.1722 µm	Width of the peak at the top of the soot particle bump.	Aerodynamic simulations to understand droplet behavior.
Mid Width of Bump	1.4 µm	Width at the midpoint of the bump.	Structural stability and water flow analysis.
Peak Radius	3.2 µm	Radius at the top of the soot particle bump.	Analysis of pressure distribution and droplet retention.
Water Harvesting Rate (Hydrophilic SLIPS)	326.4 mg/cm2^22/h	Rate of water collection on hydrophilic surfaces.	Simulation of droplet coalescence and transport efficiency.
Water Harvesting Rate (Fluorinated SLIPS)	191.6 mg/cm2^22/h	Rate of water collection on fluorinated surfaces.	Droplet movement and collection simulations.
Contact Angle (Superhydrophobic surface)	160°	Contact angle on superhydrophobic surfaces.	CAD design for surface treatment.
Contact Angle (Hydrophilic surface)	75°	Contact angle on hydrophilic surfaces.	FEA simulation to analyze droplet pinning and movement.
Critical Droplet Diameter	Dc = function of θAdv	Diameter at which droplets detach from the surface.	Simulating gravitational forces on droplet movement.

Manufacturing Process and Material

Process	Description
Materials	Candle soot particles, RTV-1 binder, lubricants (hydroxyl-terminated PDMS, methyl-terminated PDMS, perfluorinated polyether).
Substrates	Glass slides and silicon wafers were used as substrates.
Deposition Method	Soot particles were deposited using a candle flame onto RTV-coated substrates and infused with lubricants.
Characterization Tools	SEM, AFM, and ESEM were used for surface analysis and condensation behavior observation.

Reference:
https://drive.google.com/file/d/1W5qdulVna-ViVcg_yVE9k-ZCZwda1QuQ/view?usp=sharing

5. Fog Harvesting Devices Inspired from Single to Multiple Creatures: Current Progress and Future Perspective [5]

Numerical data and design parameters

Parameter	Value/Description	Unit
Thermal conductivity (Cu-SHB-SHL fabric)	0.084 to 377	W/m·K
Optimal SHB/SHL ratio	01:03	Ratio
Fog harvesting efficiency (Cu-SHL-SHB)	1432.7	$mg/h·cm^2$
Fog harvesting efficiency (spray method)	1707.25	$mg/h·cm^2$
Maximum fog harvesting efficiency (UV-exposed SHB/SHL)	244.7	$mg/h·cm^2$
Cactus-inspired spine tip angle	15° (optimal Laplace force)	Degrees
Water transport efficiency (steel needle Janus structure)	21,000	$mg/h·cm^2$
Distance between spider silk spindle knots	1.43 - 1.67	mm
Spider silk fog harvesting efficiency	4230 - 4570	mg/h
Nepenthes-inspired SLIPS initial fog harvesting	5440	$mg/h·cm^2$
Nepenthes-inspired SLIPS after 7 days	5380	$mg/h·cm^2$

Manufacturing processes and materials

Component	Material	Manufacturing Process Description
SHB-SHL Fabric	Polyester fabric, Cu NPs	Textile weaving followed by copper nanoparticle deposition
Cactus Spine	3D-printed plastic	Layer-based immersion surface stacking and kirigami cuts
Spider Silk Fibers	Chitosan, CaCl2	Electrostatic spinning and fluid coating
SLIPS (Slippery Liquid-Infused Porous Surface)	ZnO nanorods, PDMS	UV illumination grafting, solidification via silicone oil impregnation
Janus Foam	SiO2, melamine foam	Needle insertion and structural modification for water transport
Hybrid membrane (CNN/PNF)	Cu mesh, PNF, Cu (OH)2 nanoneedles	Electrospinning and anodic growth on mesh for hybrid wettability system
HOMC (Hierarchical Origami Moisture Collector)	Plastic films, metal inserts	Femtosecond laser etching, origami folding, and surface structuring
Asymmetric Fibers	Polymer materials, SHB/SHL pattern	Droplet microfluidics combined with 3D printing for asymmetric fiber design

Reference:
https://drive.google.com/file/d/1ONmPi6ybnVr3_wpEzNQO8VPvzw1MJ6pT/view?usp=sharing

6. Facile and Large-Scale Fabrication of a Cactus-Inspired Continuous Fog Collector [6]

Numerical data and design parameters

Parameter	Value/Description	Unit
Micro-tip height	1.7	mm
The apex angle of micro-tips	14°	Degrees
Tip-base width ratio (PDMS= 2:1)	0.52 ± 0.01	-
Micro-tip height (density effect)	1.22 to 2.76	mm
Water collection rate (single tip)	07-Sep	μg/s
Water collection rate (front micro-tip)	>15	μg/s
Water collection rate (rear micro-tip)	~5	μg/s
Fog velocity for testing	45-50	cm/s
Water collected at a velocity of 45-50 cm/s	3	mL/10 min
Optimal PDMS ratio	02:01	-
Fog jet distance from collector	3	cm
Maximum distance between micro-tip ridges	04-May	mm

Manufacturing process and materials

Component	Material	Manufacturing Process Description
Micro-tip array	PDMS, Cobalt Magnetic Particles (MPs)	Spin-coating PDMS-MP mixture on patterned polystyrene with magnetic field alignment
Polystyrene substrate (patterned)	Polystyrene	Mechanical process to create geometric patterns (0.5 x 0.5 x 0.13 mm pits)
Hydrophilic matrix for water absorption	Cotton fabric	Integrated with the micro-tip array for water absorption and transport
Magnetic field intensity	Neodymium magnet (0.5 Tesla)	Used to align MP arrays during spin-coating for ordered micro-tip formation
PDMS-MP mixture ratio	2:1 (optimal)	Adjusted for tip morphology and fog collection efficiency

Reference:
https://drive.google.com/file/d/1E-1DFH_DoHgSzGutomdwHqmdiliM9ktf/view?usp=sharing

7. 3D Bionic Water Harvesting System for Efficient Fog Capturing and Transporting [7]

Numerical data and design parameters

Parameter	Value/Description	Unit
Fog harvesting rate	0.5027	$g\ cm^2\ min^{-1}$
Wire diameter (optimal)	0.2	mm
Wire diameter range tested	0.05 - 0.40	mm
Sawtooth spacing (optimal)	1	mm
Spacing range tested	0.50 - 2.00	mm
Optimal frame inner size	36 × 36	mm
Copper frame thickness	0.3	mm
Tilt angle (optimal)	75	Degrees
Maximum fog harvesting efficiency improvement	180% (compared to individual monolithic fog collectors)	%
Water harvested in 8 minutes	37.5	mL

Manufacturing Process and Materials

Component	Material	Manufacturing Process Description
Copper wires	Pure copper (99.99%)	Wound around copper sheet frames to form spatial triangles
Copper frame	Pure copper (99.99%)	Laser-cut with serrated edges, cleaned via ultrasonic treatment (acetone, ethanol, deionized water)
Solid slip surface treatment	Paraffin wax	Applied to copper frame after superhydrophobic surface treatment, vacuum dried for 10 hours
superhydrophobic (SHL) surface	Copper surface treated with NaOH and $(NH_4)_2S_2O_8$	Chemical etching to enhance droplet capture
Superhydrophobic (SHB) surface	Octadecanethiol (ODT)-modified copper surface	Grafted to copper surface after SHL treatment
3D-printed storage system	Polylactic acid (PLA)	Designed with SolidWorks, printed to create enclosed or open water storage systems
Laser etching process	Copper sheet with serrations	Power 90%, frequency 90 kHz, 20-minute etching process

Reference:
https://drive.google.com/file/d/1xvx_YEkkuQYwN7nyTRBLy8mFnAilzpXM/view?usp=sharing

8. **Drops on a Conical Wire** [8]

Numerical data and design parameters

Parameter/Dimension	Value	Description
Initial Wire Radius (r_0)	1 mm	Starting radius of the copper wire before etching
Withdrawal Speed (V)	1.5 mm/s	The speed at which the wire is withdrawn from nitric acid.
Acid Concentration	60%	Nitric acid concentration used for etching.
Radius Gradient (dr/dz)	4×10^{-3} to 15×10^{-3}	Range of conical wire radius gradients produced.
Drop Volume (Ω)	0.1 mm^3 to 3 mm^3	Volumes of silicone oil drops were used for testing.
Drop Surface Tension (γ)	19.7 mN/m	Surface tension of the silicone oil used in the experiments.
Drop Density (ρ)	910 kg/m^3	Density of the silicone oil.
Film Thickness (t)	2–6 μm	The thickness of the liquid film after coating.
Film Withdrawal Speed (V_0)	0.15 cm/s	The speed at which the liquid film is deposited on the wire.
Laplace Pressure Formula	$\Delta P = 2\gamma / (r + h)$	Governs the pressure gradient driving the droplet.
Equilibrium Tilt Angle (β)	Varied with wire radius	Measured to determine the balance of forces acting on the drop.
Viscosity Range (η)	5–100 mPa·s	Range of silicone oil viscosities tested to study motion dynamics.
Capillary Length (κ^{-1})	$\sqrt{(\gamma / \rho g)}$	Describes the balance of capillary effects and gravity.

Reference:
https://drive.google.com/file/d/14I_f81uAhB7gE3bsxZ7jBbF-qeAO0wWA/view?usp=sharing

9. Three-dimensionally structured Flexible Fog Harvesting Surfaces Inspired by Namib Desert Beetles [9]

Numerical data and design parameters

Parameter	Value/Condition	Description/Role	Purpose in CAD/FEA Simulation
Surface Geometry	3D Pyramidal Bumps (210 µm height)	Bumps are patterned on the PDMS background surface.	Simulates the vapor diffusion and fog collection efficiency on a 3D surface.
Pattern Spacing	1.25 mm	Distance between adjacent bumps on the surface.	Affects droplet coalescence and distribution across the surface.
Surface Area	2 cm x 2 cm	Total area of the fog-collecting surface.	Defines the surface size for analyzing water collection efficiency.
Sample Tilting Angles	45°, 67.5°, 90°	Different angles for testing fog collection efficiency.	Simulates how tilt affects water collection efficiency in FEA.
Super hydrophilic CuO Region Contact Angle	<5°	The contact angle for water on CuO regions.	Simulates droplet nucleation, growth, and movement.
Hydrophobic PDMS Region Contact Angle	135°	The contact angle for water on PDMS regions.	Simulates how droplets roll off the hydrophobic PDMS background.
Diffusion Layer Thickness	10 mm	Thickness of the depletion layer in the FEA model.	Represents vapor concentration layer for diffusion flux analysis.
Fog Stream Distance	10 cm	Distance from the fog source to the surface.	Used to simulate the real-world interaction of fog with the surface.
Water Collection Rate (WCR)	16x higher on curved 3D surfaces	Water collected per hour on curved 3D surfaces vs. flat designs.	Evaluates water collection efficiency differences between designs.
Boundary Conditions	Fog Stream in Parallel	The fog stream runs parallel to the surface.	Simulates adverse fog stream conditions to test surface efficiency.
Von Mises Stress	Peaks on PDMS Background	Stress distribution peaks in the hydrophobic PDMS region.	Simulates mechanical performance of curved surfaces under stress.
Material Properties (PDMS)	Hydrophobic (Contact Angle: 135°)	A hydrophobic background repels water droplets.	Simulates droplet behavior on

			hydrophobic PDMS regions.
Material Properties (CuO Bumps)	superhydrophobic (Contact Angle: <5°)	CuO bumps attract and coalesce water droplets.	Simulates droplet nucleation and coalescence on CuO regions.

Manufacturing Process and Materials

Process	Description
Materials Used	Copper (Cu), Copper Oxide (CuO), Polydimethylsiloxane (PDMS).
Cu Layer Thickness	95 µm (square pattern) and 25 µm (surrounding region).
Oxidation Process	CuO is formed by oxidation in an alkaline solution (NaClO2, NaOH, Na3PO4·12H2O).
PDMS Molding	PDMS precursor was poured and cured on a CuO-Cu composite layer to create the hybrid surface.
Etching Process	Timed etching with Cu etchant to create 3D patterns.

Reference:
https://drive.google.com/file/d/1_NitTEa94KBbs6X8kES8Isa9kgh9Ez9j/view?usp=sharing

10. Structured cone arrays for continuous and effective collection of micron-sized oil droplets from water [10]

Numerical data and design parameters

Parameter	Value/Condition	Description	Application in CAD/FEA Simulation
Needle Height	500 µm	Height of the conical needles.	Used for creating the 3D structure in CAD and stress analysis in FEA.
Base Diameter	0.5 mm	The diameter of the needle base is made from copper wire.	Defines the base for FEA structural integrity simulations.
Conical-Apex Angle	10°, 15°	The apex angles of the conical needles were tested.	Simulates the effect of the apex angle on droplet collection efficiency.
Needle Separation	0.25 mm - 0.5 mm	Distance between conical needles in arrays.	Impacts droplet coalescence and collection rate in the FEA simulations.
Needle Material	Copper (Cu) and PDMS	Materials used for fabricating the needles (Cu and PDMS).	Material properties for CAD and FEA, including wettability and stiffness.
Oil Droplet Size	1-10 µm	Size of the oil droplets collected by the needles.	Simulates droplet behavior and dynamics in FEA.
Droplet Motion Speed	2 mm/s	The speed at which droplets are driven towards the needle base.	Models' fluid flow and droplet dynamics in FEA.
Roughness of Needles	0-6.7 µm nanorods	Length of nanorods on rough needles to enhance oleophilicity.	Used to evaluate surface roughness impact on droplet collection.
Throughput Rate	1.128 $mL/cm^2/s$	The volume of oil-water mixture processed by the array per second per cm^2.	Simulates efficiency in large-scale separation systems.
Oil Collection Efficiency	93.7-98.9%	Efficiency of oil collection on smooth and rough needle arrays.	Used as a performance metric in FEA for system optimization.
SEM Characterization	Used for surface morphology	Scanning Electron Microscopy (SEM) images of needle surfaces.	Provides structural details for CAD modeling of surface features.

Manufacturing Process and Materials

Step	Description
Needle Fabrication (Cu Needles)	Electroetching of commercial Cu wire in a CuSO4_44 solution, then roughened in NaOH and $Na_2S_2O_8$.
Needle Fabrication (PDMS)	Commercial sewing needles are used to form molds in polyethylene (PE), then replicated using PDMS.
Surface Roughening	Nanorods are added to the needle surface by varying roughness through controlled etching.
Weight Measurement Method	Needles suspended on micro-electromechanical balance systems to measure oil collection weight changes.
Experimental Setup	The oil-water mixture was sprayed onto conical needles immersed in water, oil was collected via coalescence and motion.

Reference:
https://drive.google.com/file/d/1WxDyIvGIqjPjmSKub9w51EFLWDQBi1FW/view?usp=sharing

11. Bioinspired asymmetric amphiphilic surface for triboelectric enhanced efficient water harvesting [11]

Numerical data and design parameters

Parameter	Value/Condition	Description	Application in CAD/FEA Simulation
Surface Geometry	Asymmetric Spines, 10 mm height	The structure consists of bioinspired asymmetric spines for water harvesting.	Used for 3D modeling and simulating droplet behavior and water collection.
Base Width	1 mm	The base width of each spine.	Affects structural stability in FEA and surface area in water collection.
Angle of Spine and Channel	30°, 60°, 90°	Various angles between the spine and hydrophobic channel were tested.	Simulates droplet transport at different angles to optimize the design.
Spacing Between Spines	2 mm (ipsilateral), 1.2 mm (contralateral)	Vertical distance between spines on each side of the spine structure.	Influences droplet coalescence and speed of water shedding in FEA.
Material (Hydrophobic Channel)	Fluorinated ethylene propylene (FEP)	The hydrophobic material is used to enhance droplet transport.	Simulates material interaction with droplets and electrostatic effects.
Water Harvesting Rate (WCR)	93.18 $kg/m^2/h$	Achieved optimal water collection efficiency.	Used to validate FEA simulation for water transport and collection.
Triboelectric Potential	103.2 V	Open-circuit voltage generated by triboelectric effect during water collection.	Simulates energy generation from water droplets.
Droplet Size	Average 96 µL	The volume of droplets collected.	Simulates droplet coalescence and behavior on the surface.
Electrode Size (Upper)	1 mm x 6 mm	Size of copper foil upper electrode for droplet triboelectric nanogenerator.	Used in FEA simulations for energy generation from droplets.
Contact Angle (Hydrophobic Channel)	108° (advancing), 61° (receding)	Contact angles on the FEP surface were measured for water droplets.	Models' droplet movement and slippage on hydrophobic surfaces in FEA.
Contact Angle Hysteresis	47°	Difference between advancing and receding contact angles.	Affects droplet pinning and transport,

			considered in FEA simulations.
External Charge for Electrostatic Adsorption	90.6 V	The voltage is applied to enhance the electrostatic adsorption of water droplets.	Simulates how external charges improve water collection efficiency.

Manufacturing Process and Materials

Process	Description
Material Used	Fluorinated ethylene propylene (FEP) film, amphiphilic cellulose ester coating (ACEC), copper electrodes.
Spine Fabrication	Laser engraving of the FEP membrane to form cactus-like spine structures with asymmetric designs.
ACEC Coating Process	ACEC is applied on the FEP spines by dip coating to introduce hydrophilic domains for droplet nucleation.
Triboelectric Nanogenerator Fabrication	Copper foil electrodes are placed over FEP to generate electricity from droplet impact.
Surface Characterization	SEM and industrial microscopy were used to verify spine morphology and surface roughness.
Electrical Measurement	Open-circuit voltage and triboelectric charge were measured using an electrometer.

Reference:
https://drive.google.com/file/d/1qKyayLTTfHFgXsCWVTNL465T2LZZlfhT/view?usp=sharing

12. Structural Design of Efficient Fog Collectors: A review [12]

Numerical data and design parameters

Parameter/Dimension	Value	Description/Source
Fog Water Yield	1–10 L/m^2/day	Reported range of fog water yield from previous literature
Surface Area of SFC	1 × 1 m^2	Standard Fog Collector (SFC) used in commercial studies
Surface Area of LFC	40–50 m^2	Large Fog Collector (LFC) dimensions
Mesh Solidity (s)	35%–40%	Recommended solid fraction for mesh efficiency
Polypropylene Fiber	0.1 mm thickness, 1 mm width, 10 mm pore size	Mesh structure dimensions
Wind Speed	3.5 m/s	Optimum wind speed for fog collection
Liquid Water Content (LWC)	0.22 to 0.73 g/m^3	LWC range in camanchaca fog
Mesh Efficiency (Me)	26%–69%	Mesh efficiency for droplet sizes of ~15 μm
Capture Efficiency (ηcapt)	Up to 95%	Efficiency for droplets of 11-15 μm at wind speeds of 8 m/s
Stokes Number (Stk)	Stk ≫ 1	A higher Stokes number leads to better droplet interception
Pressure Drop Coefficient (C_0)	Variable	Affects fog collector aerodynamic collection efficiency
Drag Coefficient (Cd)	Variable	Influences the wind flow through the mesh
Optimal Solidity for Double-Layer Mesh	0.50–0.75	Considered best for maximizing aerodynamic collection efficiency
Raschel Mesh	38% Solidity per layer	Commonly used for fog harvesting
Copper Wire Diameter	1 mm	Used in periodic conical copper wire (PCCW) for fog collection
Water Collection Efficiency	6 L/m^2/day	Maximum water collection efficiency

Reference:
https://www.sciencedirect.com/science/article/abs/pii/S2352186420314693?via%3Dihub

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